

Robot Design Exercise

CONTEXT

The Setup

Students have just seen Orbit Odyssey rules for the first time. They know the game: collect ping pong balls, score them in the goal. They have NOT yet seen the robot hardware or the codebase subsystem list.

PHASE 1 — PRESENT THE GAME (1–2 min)

- Show the Orbit Odyssey rules page on the projector (or read aloud): robots collect ping pong balls and score them.
- Put the physical XRP robot on the table where everyone can see it. Don't name the mechanisms yet.
- Say: "Before I show you the code, I want you to think like an engineer. What would this robot need to do?"

PHASE 2 — STUDENT BRAINSTORM (4–5 min)

Ask: "What actions does this robot need to perform to win the game?"

Ask: "If you had to break those actions into separate jobs, what would they be?"

Ask: "What hardware would each job use — motors, servos, sensors?"

INSTRUCTOR NOTE Let students call out answers freely. Write them on the whiteboard as-is — don't correct yet. Aim for 3–5 ideas.

PHASE 3 — SOCRATIC NARROWING (2–3 min)

Q: "Does moving the robot feel different from lifting the elevator?"

→ Drive vs. mechanism — two separate jobs, two separate subsystems.

Q: "What if the arm breaks — should it take down driving too?"

→ Isolation principle: subsystems fail independently.

Q: "The elevator goes up and down. The scoop tips. Same subsystem or different?"

→ Different hardware, different jobs → different subsystems.

REVEAL — "Here are the four subsystems on your XRP robot."

Drive

Motors 0 & 1
Encoders + Gyro

Move the robot.
Arcade drive IK.

Arm

XRPServo 4
(Servo 1)

Angle-controlled
rear arm.

Elevator

XRPMotor 4
Encoder

Lift mechanism.
Volt-controlled.

Scoop

XRPServo 5
(Servo 2)

Dump/carry
ping pong balls.

"In Lesson 3, you'll open each of these subsystems in the code and start building one yourself."